

Python Básico

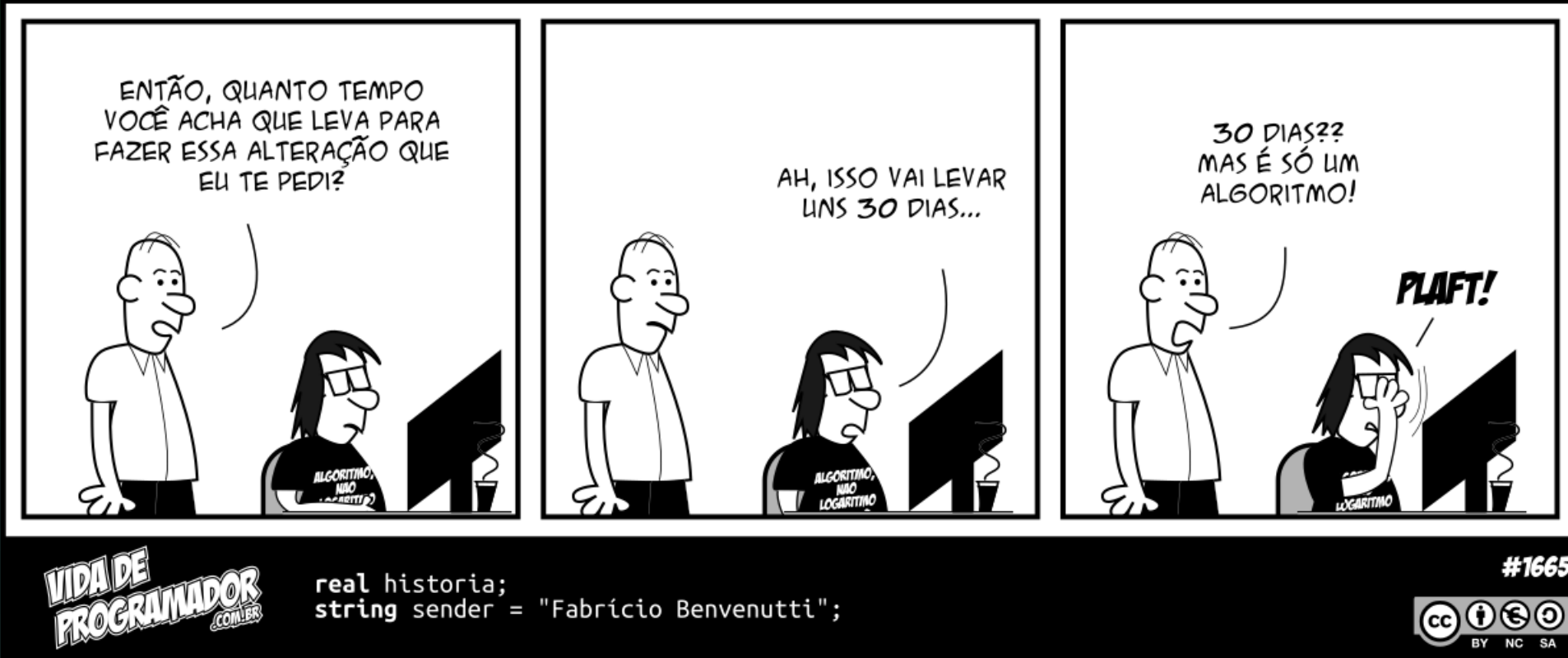
I. Algoritmo

Introdução a algoritmos 1 (conceito e história)

Guilherme Ferreira Silveira

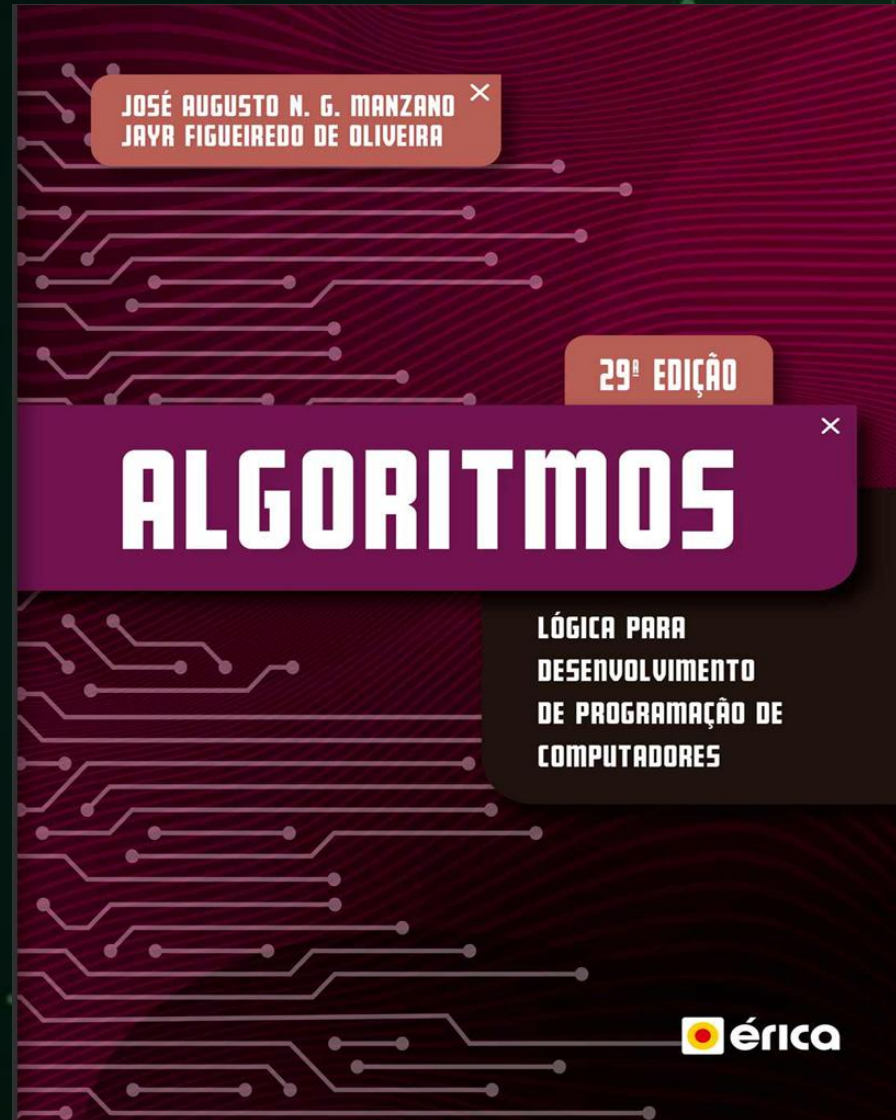
20 de março de 2026

gfsilveira@gmail.com



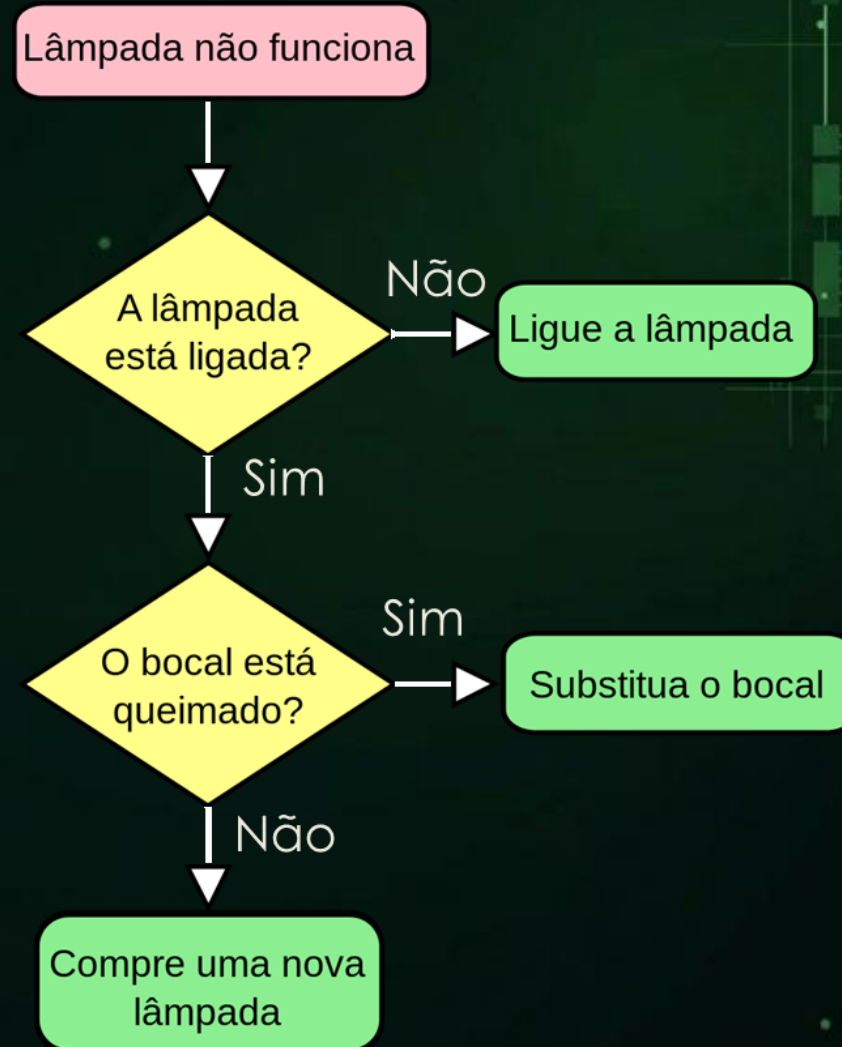
Algoritmo

Algoritmos são conjuntos de passos finitos e organizados que, quando executados, resolvem um determinado problema.



Algoritmo

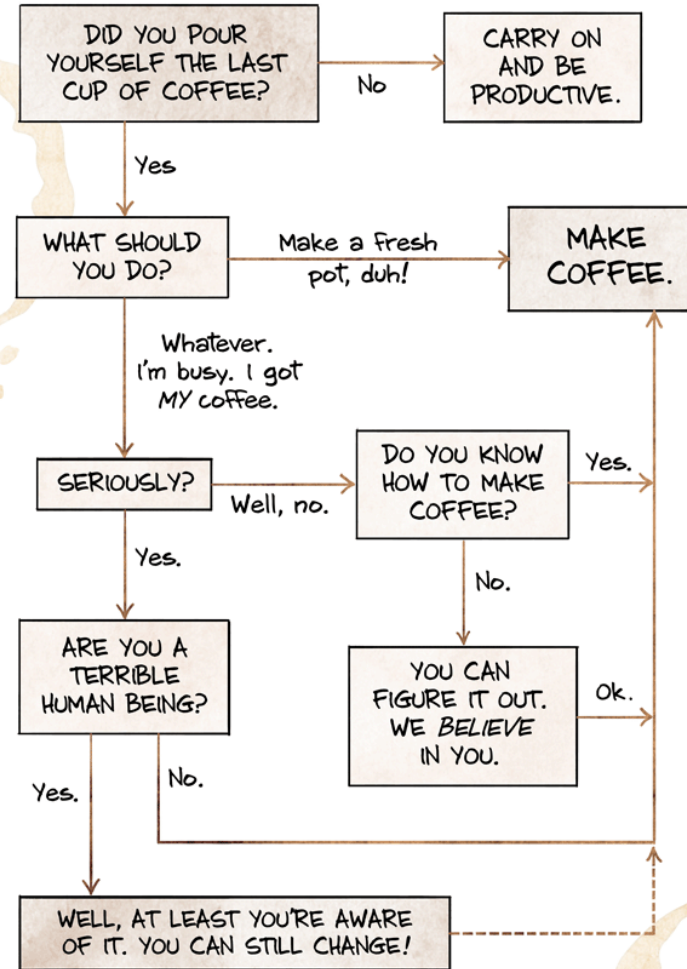
Algoritmos são conjuntos de **passos finitos** e **organizados** que, quando executados, **resolvem** um determinado **problema**.



Algoritmo

Algoritmos são conjuntos de **passos finitos** e **organizados** que, quando executados, **resolvem** um determinado **problema**.

The Office Coffee Flowchart



Maior Divisor Comum

Euclides de Alexandria – 400 a.E.C.



$$\text{MDC}(a,b) = \text{MDC}(b,r)$$

Onde:

a = número maior

b = número menor

r = resto de a/b

Como Funciona (Passo a Passo):

- 1.Divisão Inicial:** Divida o maior número pelo menor número.
- 2.Repetição:** Troque o dividendo pelo divisor anterior, e o divisor pelo resto (R). Continue dividindo..
- 3.Finalização:** Repita o processo até que o resto seja 0..
- 4.Resultado:** O MDC é o último resto diferente de zero obtido..

Algoritmos são conjuntos de **passos finitos** e **organizados** que, quando executados, **resolvem** um determinado **problema**.



Sistema numérico hindu-árabe

Abu Abdullah Muhammad bin Musa Al-Khwarizmi – 850

Persian

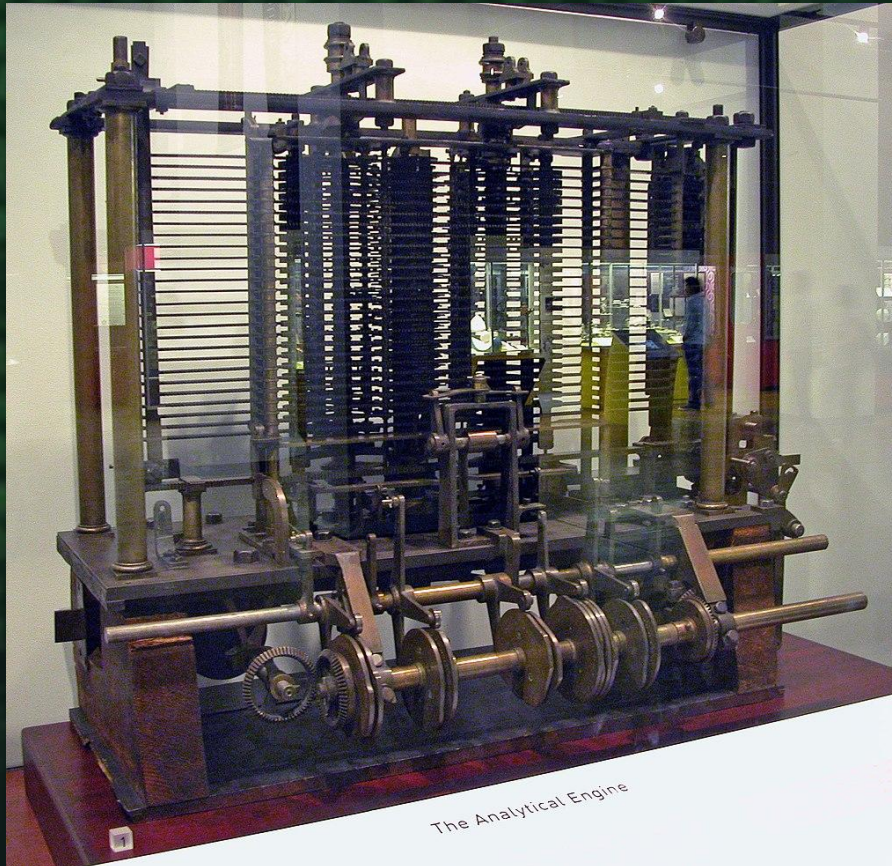
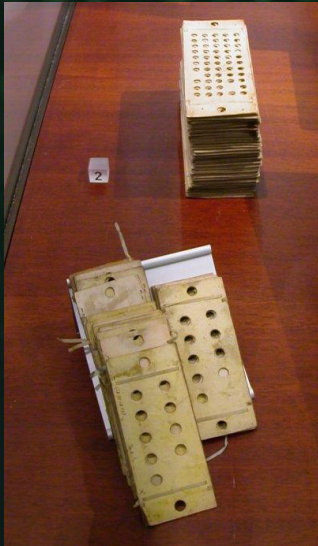
0, 1, 2, 3, 4, 5, 6, 7, 8, 9



Motor Analítico

Charles Babbage

“O pai do computador”



1833 – 1910



Primeiro Algoritmo Publicado

Ada L. Lovelace

1840, Babbage, Universidade de Turin -> Luigi Menabrea, Frances

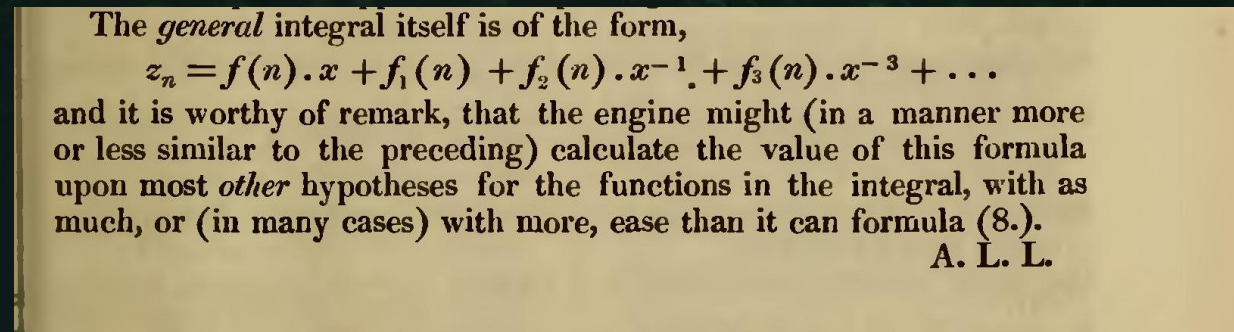
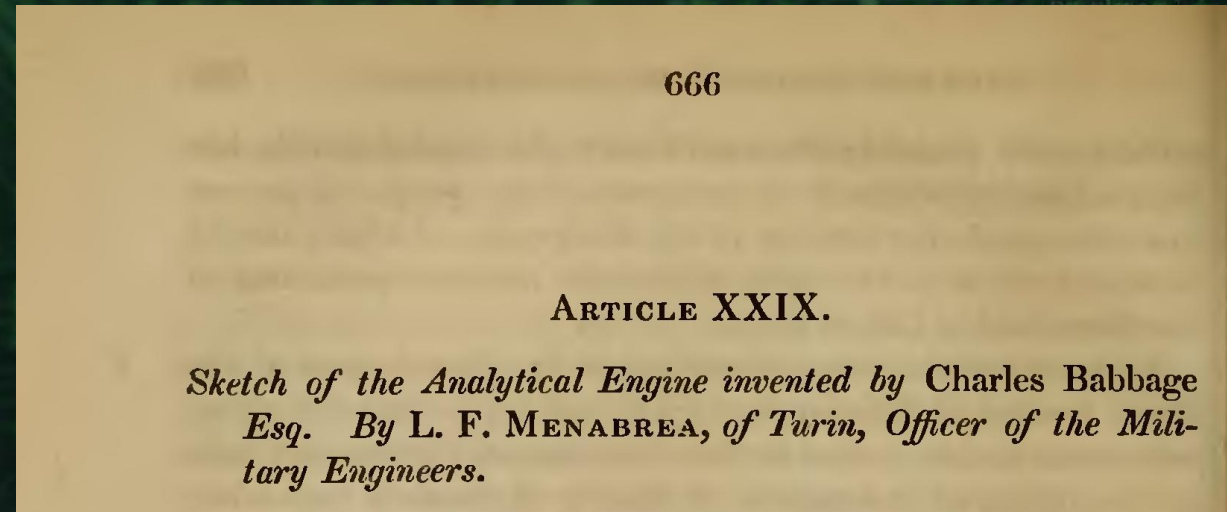
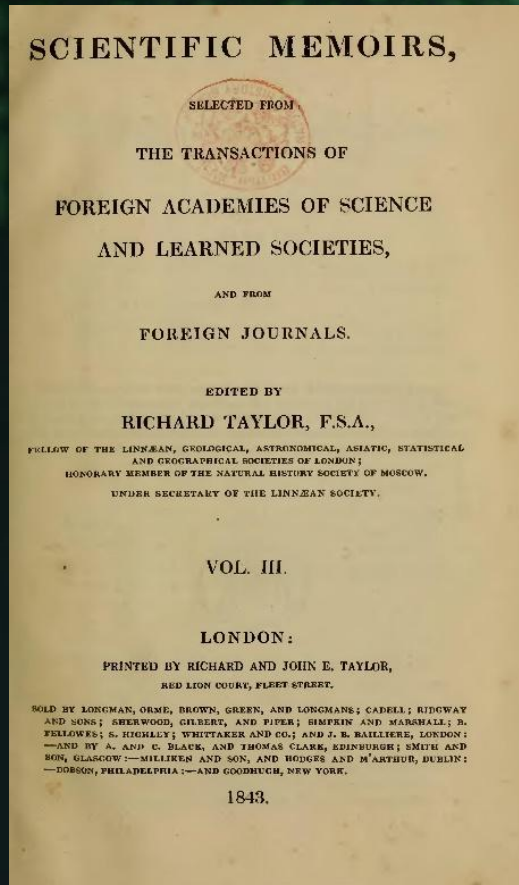
1842, Charles Wheatstone contrata Ada L. Lovelace, Inglês.

1843, publicação.



Primeiro Algoritmo Publicado

Ada L. Lovelace



Ada L. Lovelace

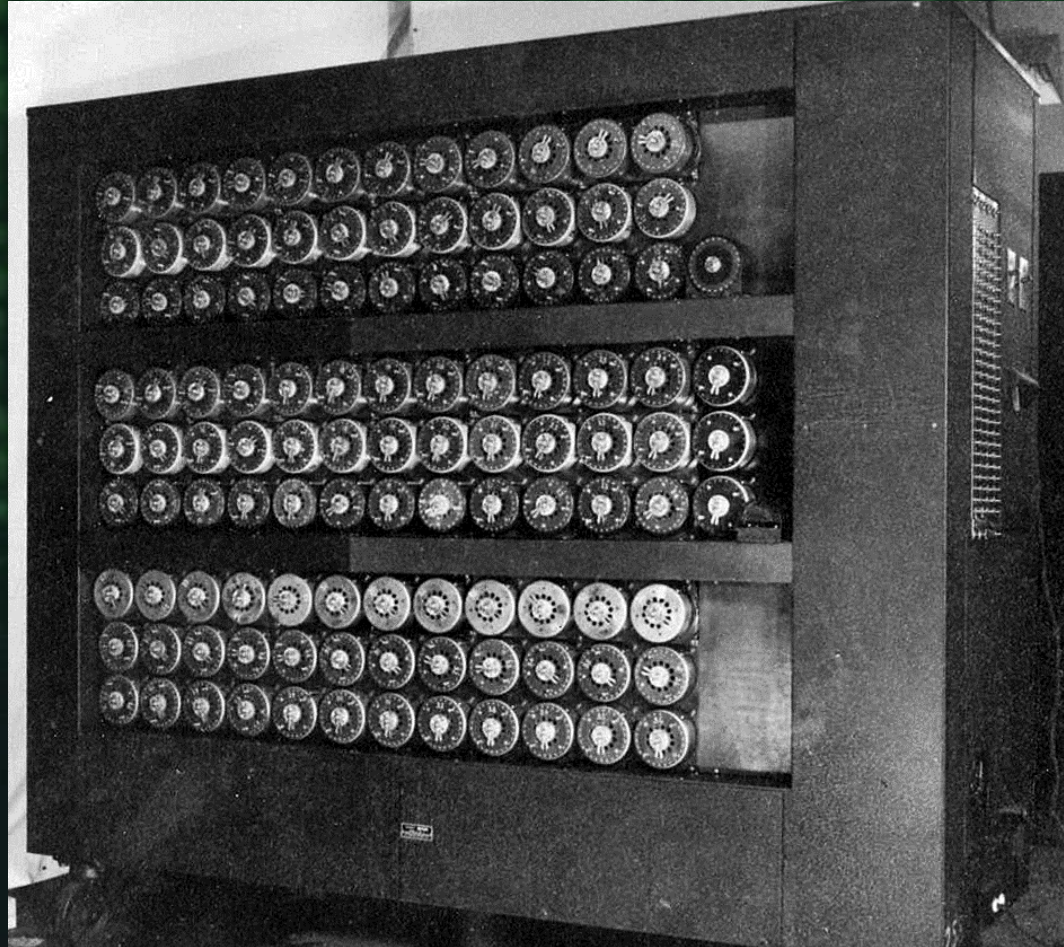
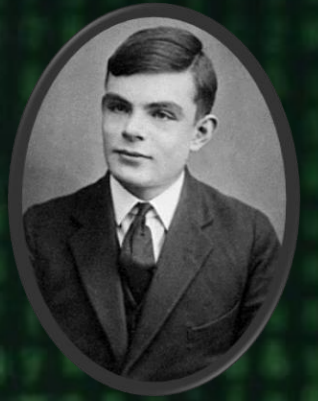
An oval-framed portrait of a woman, likely a historical figure, dressed in 18th-century fashion. She wears a dark, lace-trimmed veil or shawl over a purple and white gown. Her hair is styled in an elaborate updo with a yellow flower. She holds a quill pen in her right hand. The background is a soft, mottled brown.



Máquina de Turing

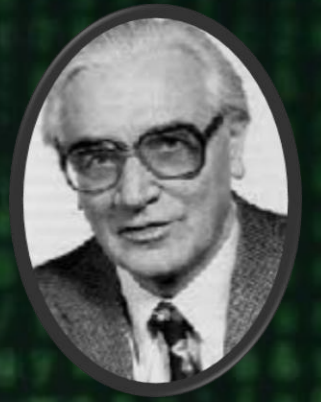
Alan Mathison Turing – 1939-40

The Bombe – 1940

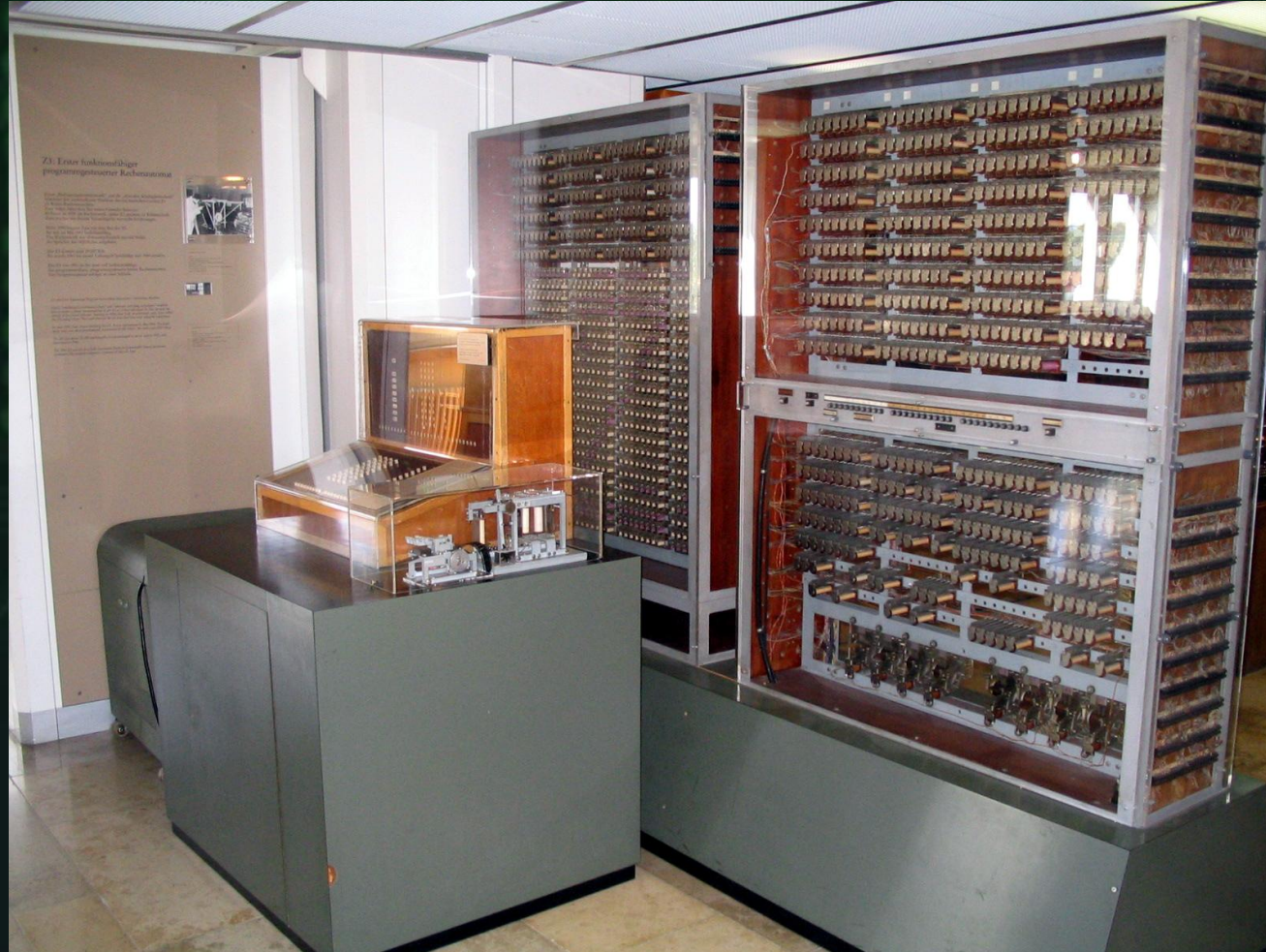


Computador Programável

Konrad Zuse



1941 – Z3



Máquina de Turing

Alan Mathison Turing – 1931



Monatshefte Math. Phys., 38 (1931), 173–198.

ON COMPUTABLE NUMBERS, WITH AN APPLICATION TO THE ENTSCHEIDUNGSPROBLEM

By A. M. TURING.

[Received 28 May, 1936.—Read 12 November, 1936.]



Máquina de Turing

Alan Mathison Turing – 1931



<http://aturingmachine.com/>, accessed in 12/03/19



Máquina de Turing

Alan Mathison Turing – 1931



<http://aturingmachine.com/>, accessed in 12/03/19



Informação = Bit

Claude Shannon – 1938

A Symbolic Analysis of Relay and Switching Circuits*

*Claude E. Shannon***

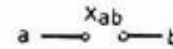


Figure 1 (left). Symbol for hindrance function

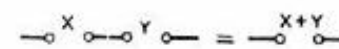


Figure 2 (right). Interpretation of addition

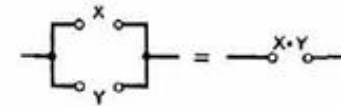


Figure 3 (middle). Interpretation of multiplication

It is evident that with the above definitions the following postulates will hold:

Postulates

1. *a.* $0 \cdot 0 = 0$
b. $1 + 1 = 1$

An open circuit in parallel with a closed circuit is a closed circuit.
An open circuit in series with an open circuit is an open circuit.
2. *a.* $1 + 0 = 0 + 1 = 1$
b. $0 \cdot 1 = 1 \cdot 0 = 0$

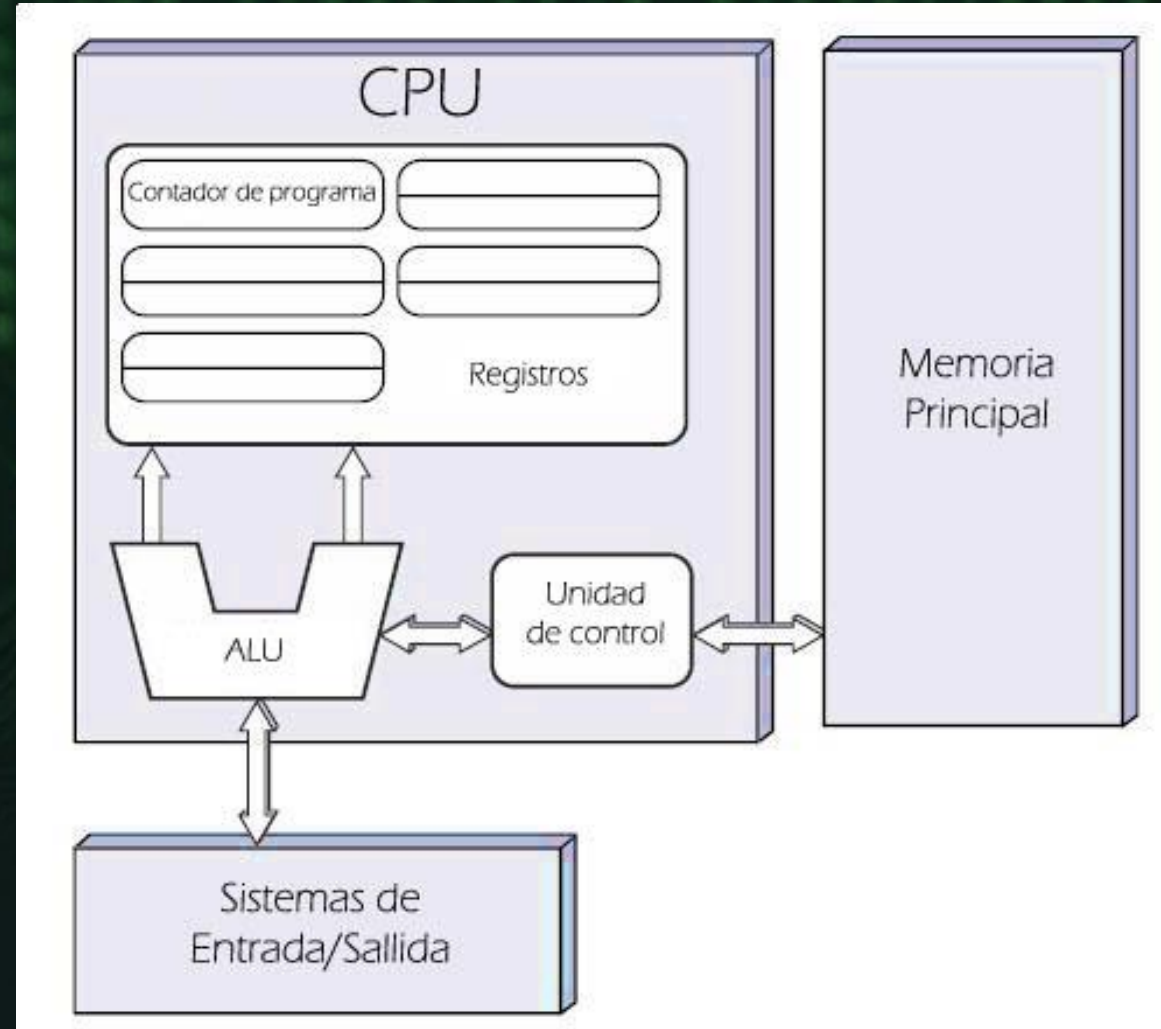
An open circuit in series with a closed circuit in either order (i.e., whether the open circuit is to the right or left of the closed circuit) is an open circuit.
A closed circuit in parallel with an open circuit in either order is a closed circuit.
3. *a.* $0 + 0 = 0$
b. $1 \cdot 1 = 1$

A closed circuit in series with a closed circuit is a closed circuit.
An open circuit in parallel with an open circuit is an open circuit.
4. At any given time either $X = 0$ or $X = 1$.



Infraestructura

John von Neumann – 1945



Programadora da Apollo 11

Margaret Hamilton

1969



Primeiro programa de computador

Contra Alm. Grace M. Hopper – 1954

UNIVAC I (UNIVersal Automatic Computer I)

FLOW-MATIC

```
(0) INPUT INVENTORY FILE-A PRICE FILE-B ; OUTPUT PRICED-INV FILE-C UNPRICED-INV  
FILE-D ; HSP D .  
(1) COMPARE PRODUCT-NO (A) WITH PRODUCT-NO (B) ; IF GREATER GO TO OPERATION 10 ;  
IF EQUAL GO TO OPERATION 5 ; OTHERWISE GO TO OPERATION 2 .  
(2) TRANSFER A TO D .  
(3) WRITE-ITEM D .  
(4) JUMP TO OPERATION 8 .  
(5) TRANSFER A TO C .  
(6) MOVE UNIT-PRICE (B) TO UNIT-PRICE (C) .  
(7) WRITE-ITEM C .  
(8) READ-ITEM A ; IF END OF DATA GO TO OPERATION 14 .  
(9) JUMP TO OPERATION 1 .  
(10) READ-ITEM B ; IF END OF DATA GO TO OPERATION 12 .  
(11) JUMP TO OPERATION 1 .  
(12) SET OPERATION 9 TO GO TO OPERATION 2 .  
(13) JUMP TO OPERATION 2 .  
(14) TEST PRODUCT-NO (B) AGAINST ; IF EQUAL GO TO OPERATION 16 ;  
OTHERWISE GO TO OPERATION 15 .  
(15) REWIND B .  
(16) CLOSE-OUT FILES C ; D .  
(17) STOP . (END)
```



COmmon Business-Oriented Language - COBOL

```
ADD 1 TO x  
ADD 1, a, b TO x ROUNDED, y, z ROUNDED  
  
ADD a, b TO c  
ON SIZE ERROR  
    DISPLAY "Error"  
END-ADD  
  
ADD a TO b  
NOT SIZE ERROR  
    DISPLAY "No error"  
ON SIZE ERROR  
    DISPLAY "Error"
```



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20 de março de 2026